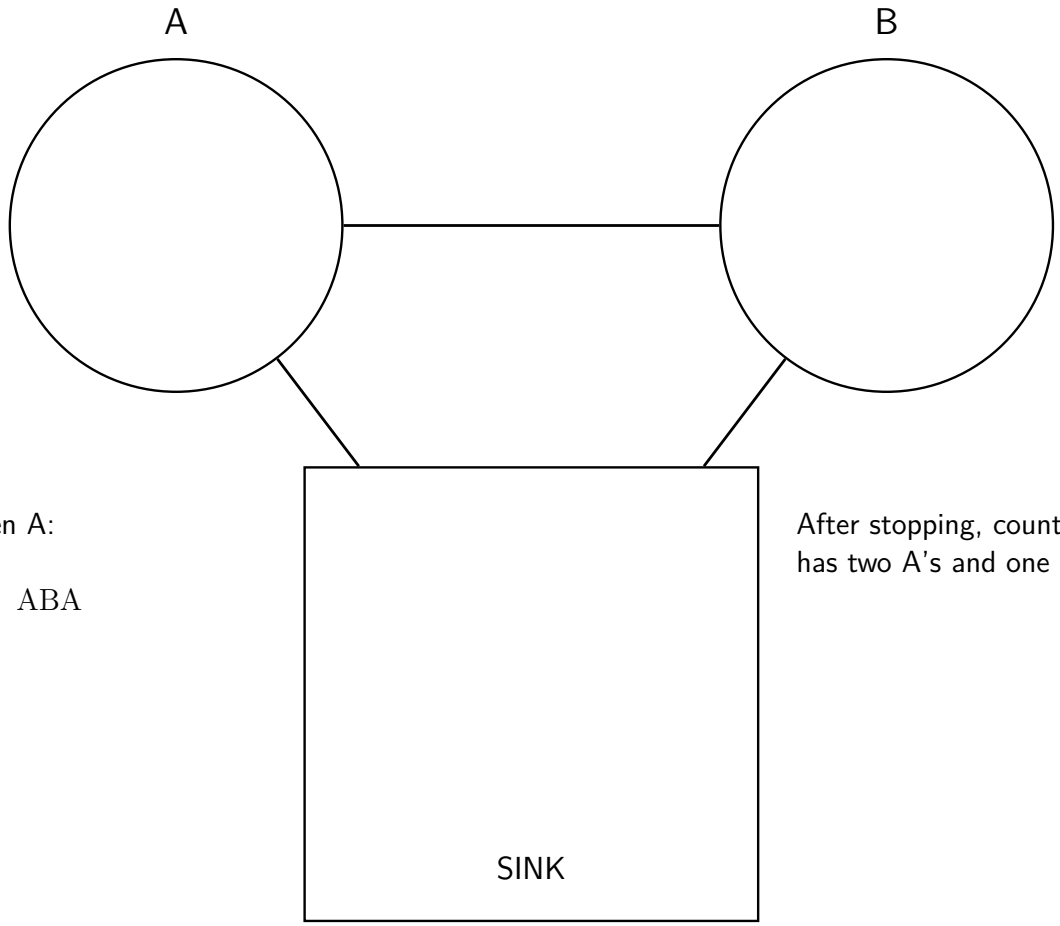
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Finish each start in two different orders, if possible. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?



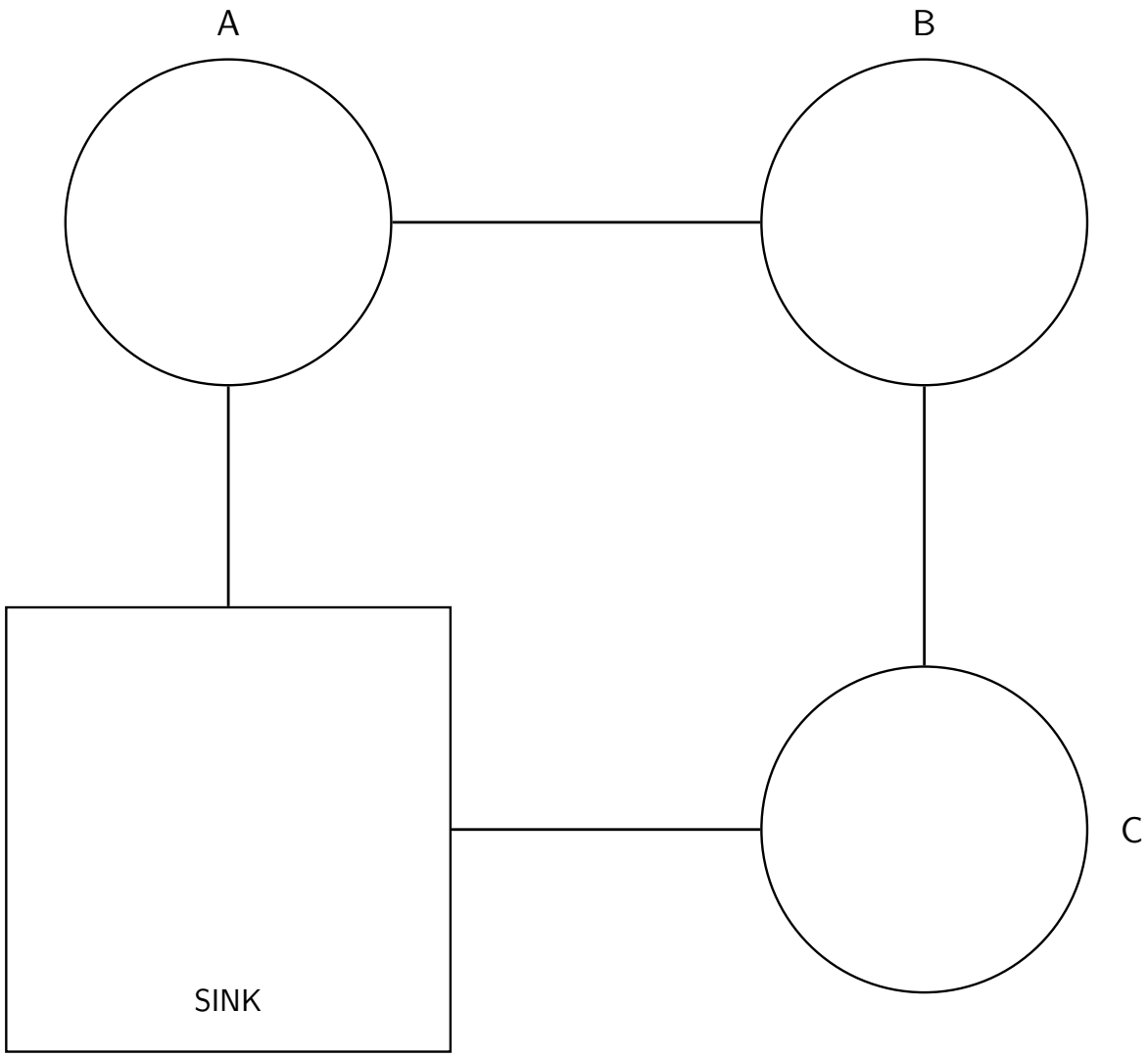
A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

After stopping, count letters: ABA has two A's and one B.

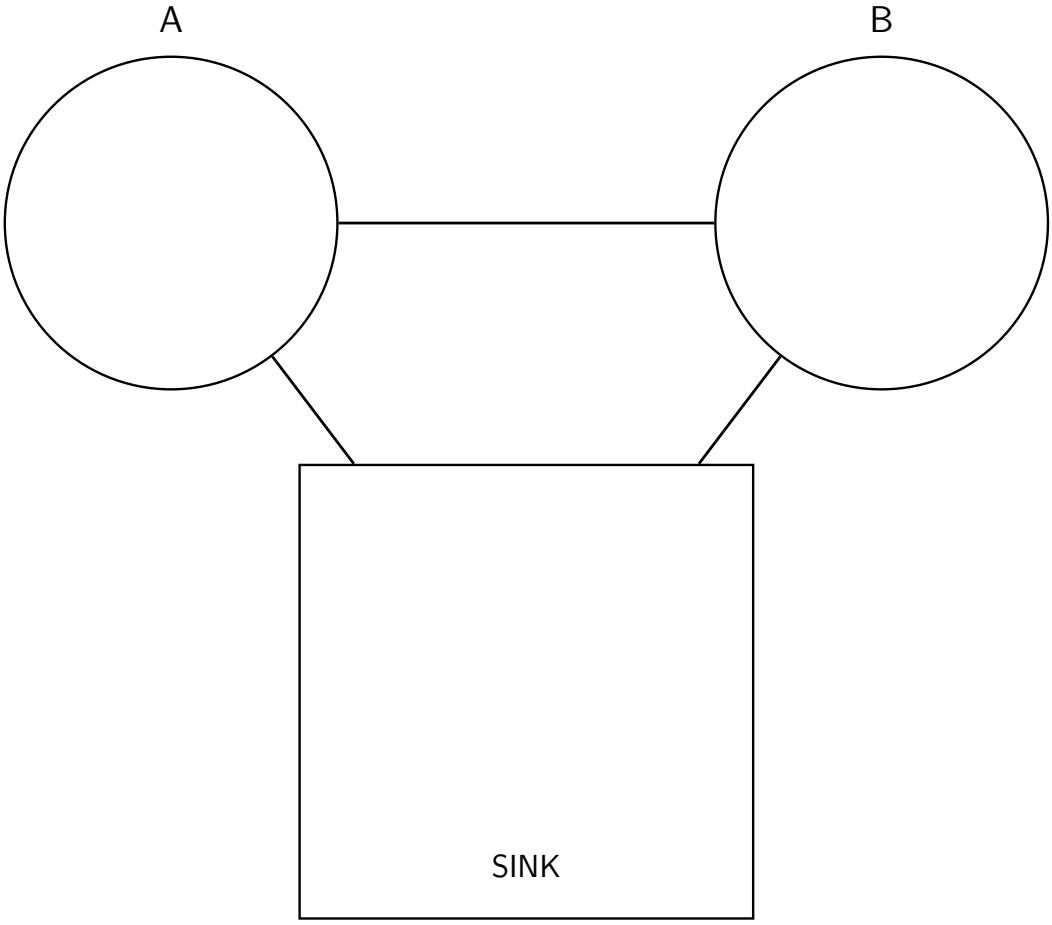
Start A	Start B	Sharing word	Finish A	Finish B
2	2			
2	2			
4	0			
4	0			
3	3			
3	3			

Problem 2: Finish each start in two different orders. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?



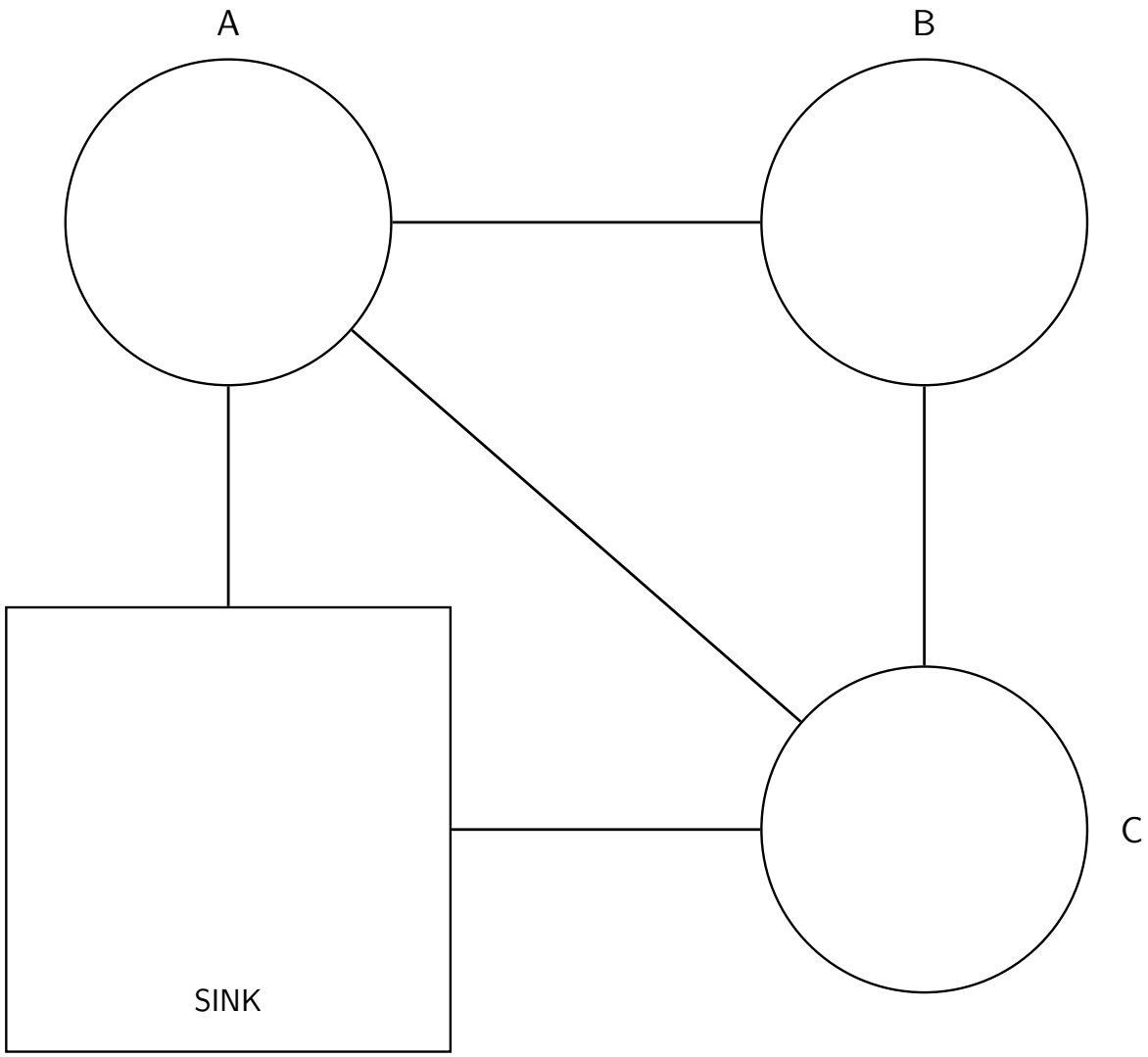
Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
0	4	0				
0	4	0				
2	1	2				
2	1	2				
2	4	2				
2	4	2				

Problem 3: Put exactly 6 chips altogether on A and B. Find every start that finishes with one chip at A and one at B. Explain how you know you have found every start that works.



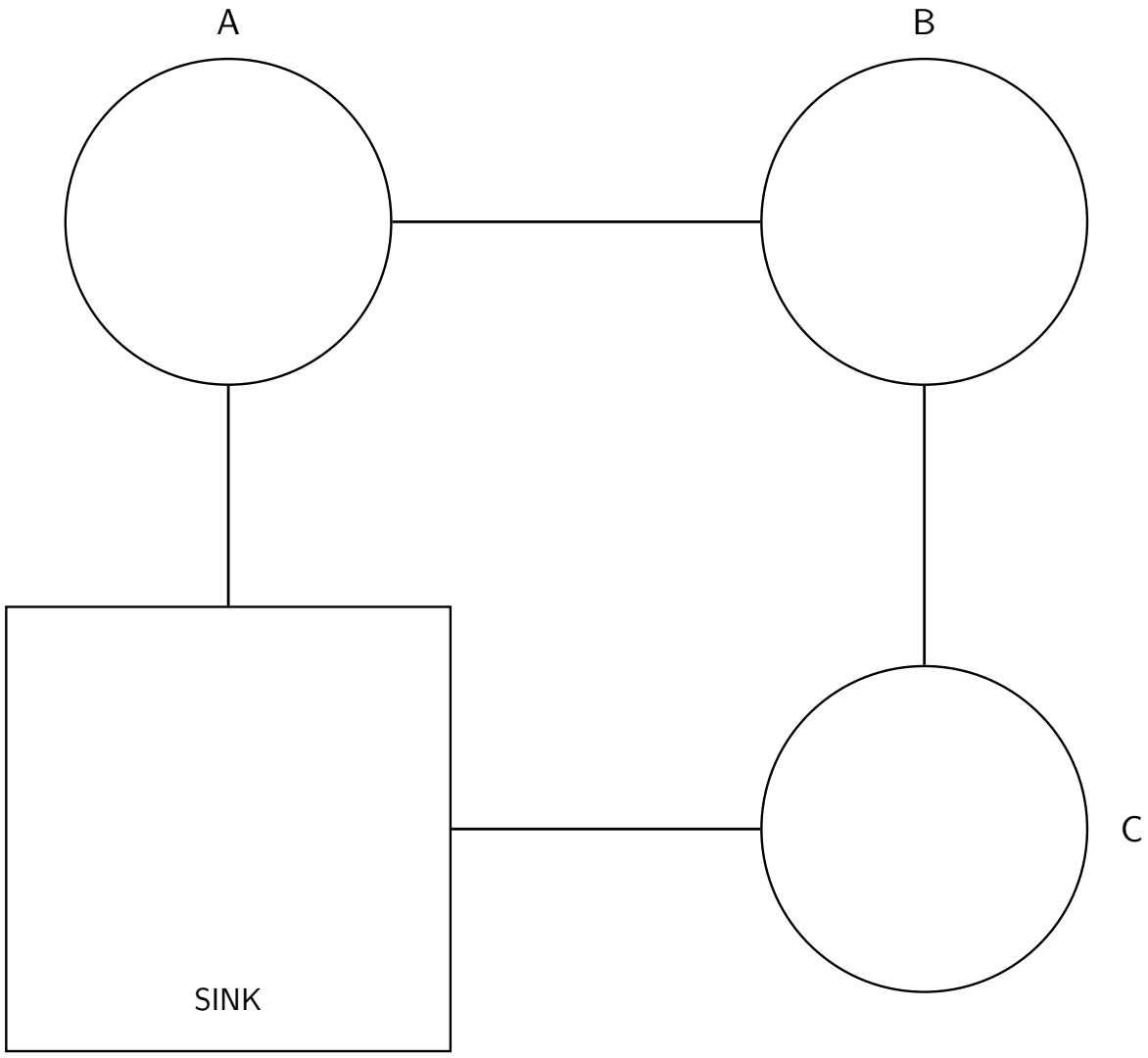
Start A	Start B	Finish A	Finish B

Problem 4: Finish each start. Find another order wherever a choice is possible. After stopping, compare the final piles and count each letter in the sharing words. Does the extra line change your conclusion about order?



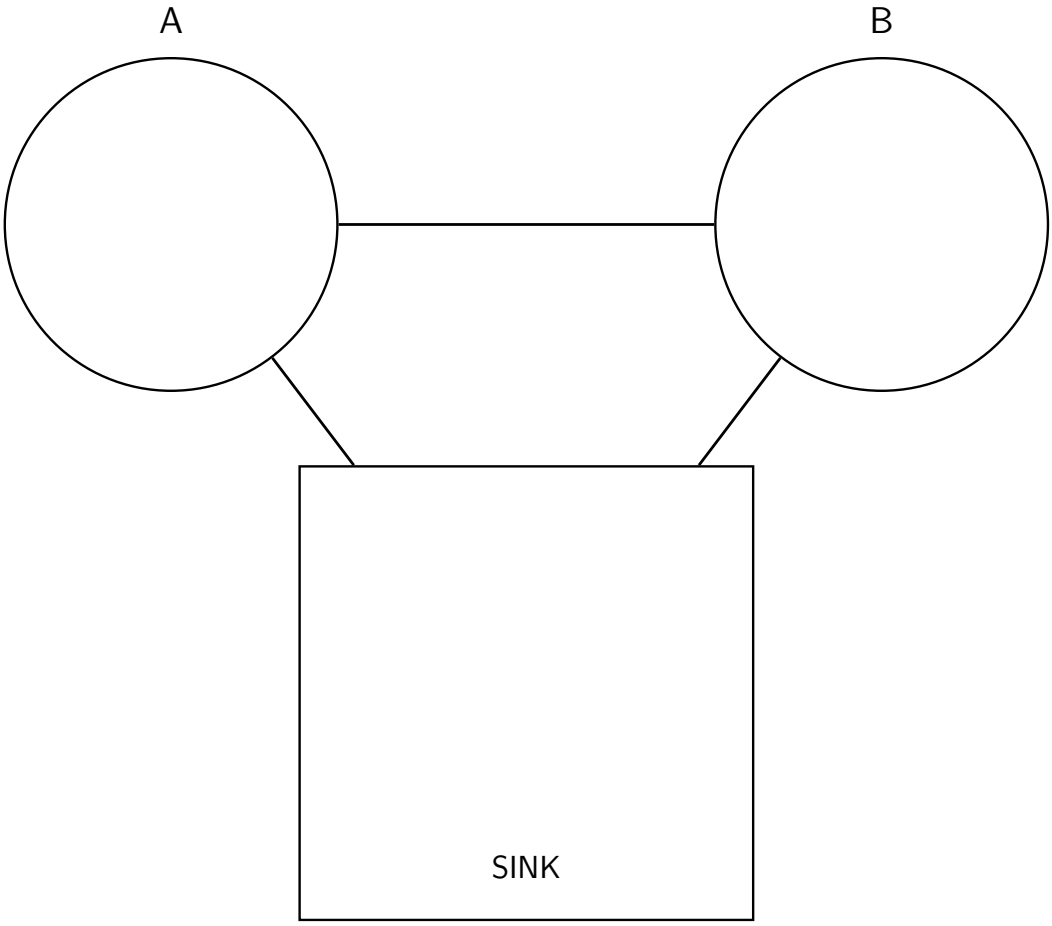
Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
3	0	3				
3	0	3				
0	6	0				
0	6	0				
2	2	2				
2	2	2				
4	1	2				
4	1	2				

Problem 5: Start empty for each trial. Add the two batches together and finish, or finish one batch before adding the other. Try both orders of the batches. Can these choices change the final piles?



Add at A	Add at B	Timing	Finish A	Finish B	Finish C
2	4	together			
2	4	2 at A first			
2	4	4 at B first			

Problem 6: Begin with empty circles. Add one chip to A and finish; keep repeating until you have added 12 chips. Begin again, adding to B instead. After at least one addition, which pairs of final piles can each rule reach?



Chips added	Add to A: final A	Add to A: final B	Add to B: final A	Add to B: final B
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				